Integrating Semantic Segmentation with Parallax Geometry for UAV-Based Structure Height Estimation

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***Abstract*—Structural measurements involving building heights are critical for analytics, disaster prevention, and infrastructure monitoring. These measurements are often acquired through traditional surveying methods, which can be inefficient and labor-intensive. Unmanned Aerial Vehicles (UAVs) provide a scalable and cost-efficient alternative. While photogrammetric techniques such as depth perception and Structure from Motion leverage UAV-based imagery, they require immense computational power that can lead to suboptimal performance in a resource-constrained context. Our research proposes the first integration of semantic segmentation with parallax geometry as an alternative to estimating building heights in a cost-efficient manner avoiding densely labeled annotations. We collected a dataset of two hundred seven manually outlined images containing oblique views of urban and residential buildings for a pre-trained deep learning model. To improve generalization, the model was fine-tuned using augmentation techniques including contrast enhancements and smoothing filters. Then, in each image, we segmented a foreground building class against a background class. Furthermore, these regions are processed by a Depth-First Search algorithm that removes extraneous fragmented regions and retains the largest contiguous building class. To test our deep learning model, we collected pairs of images with a five-meter vertical displacement using UAVs on a target building. Following parallax geometry, the vertical pixel shift across the overlapping images is used to triangulate building heights through geometric principles. The model was capable of identifying and generating masks of buildings from UAV imagery with a maximum validation Dice Coefficient of approximately ninety-six percent and a maximum validation Intersection over Union of approximately ninety-three percent. Our parallax calculations reported an accuracy of approximately ninety-three percent, excluding heavily occluded buildings in estimating the height of buildings relative to ground truth survey data. Our results present a significant advancement in photogrammetric techniques to estimate structural measurements and establish a promising foundation for cost-effective large-scale urban monitoring.**

***Index Terms*—unmanned aerial vehicles, semantic segmentation, parallax geometry, building height estimation**

I. Introduction

Unmanned Aerial Vehicles (UAVs) are frequently used in photogrammetry to capture angles of structures at distinct lines of sight due to their controllable and efficient mobility. Equipped with numerous sensors and technological capabilities, UAVs relay important information regarding their position relative to the ground. Advanced filtering systems such as the Kalman filter [1] format these metrics in a reliable manner. Additionally, their controllable flight paths yield convenient and inexpensive input on the surrounding spatial environment, allowing for the rapid acquisition of multitemporal data needed for multiple computer vision tasks. In particular, UAVs have increasingly had applications in the deep learning field, where access to hyperspectral and complex imaging is a necessity for robust aerial scene segmentation [2]. Hence, there are multiple publicly available drone datasets [2]–[7] that provide high-resolution and low-altitude images of objects needed to distinguish class objects such as structures, pedestrians, vehicles, and more.

Semantic segmentation is widely used in deep learning as a method of dividing images into meaningful classes. In particular, Fully Convolutional Networks (FCNs) and Convolutional Neural Networks (CNNs) are deep learning approaches that have seen recent success in capturing context information from UAV-based imagery [8]. Commonly used pre-trained deep learning models deriving from a CNN architecture include DeepLabV3—the model used in this study. Moreover, models trained for a specific task undergo knowledge transfer [4], a fine-tuning process which includes various data augmentation techniques such as image processing, normalization, histogram equalization, and noise reduction, all of which can improve performance and model generalizability [9].

Parallax calculations were famously used in astrophysics to progressively derive long-distance measurements such as distances between stars up to celestial bodies, leading to distance ladders [10]–[13]. Given two overlapping images and a known distance between the vertical or horizontal displacement, trigonometric calculations can be made to find missing legs in constructed triangles. These geometric principles can be applied to UAV-imagery, where the shift in an angle after a known vertical displacement in contrasting images can be used to find distances. Parallax geometry capitalizes on cost-efficiency by only requiring two distinct lines of sights, thus avoiding pixel-level annotations or depth maps—common techniques used to estimate building structures [14], [15]. To the best of our knowledge, parallax calculations have not been utilized with semantic segmentation as a means of calculating the heights of structures. This insight will be the key point of our research paper, and we summarize it with the following contributions:

- We propose a fine-tuned deep-learning model for semantic segmentation on processed UAV-based data to generate binary masks of two object classes.
- We introduce parallax geometry with semantic segmentation as a method of estimating building heights.
- We evaluate the model's accuracy across standardized metrics to ensure robust performance. Additionally, we assess the model's building heights estimation on UAV-based validation data against ground truth survey data.

The remainder of the paper is organized as follows. Section II covers recent developments in structure height estimation, drone datasets, and parallax calculations. Section III presents the architecture for our deep learning model and the novel parallax geometry. Section IV details the results and discussion regarding limitations. Finally, Section V concludes with final statements of the research.

II. Related Work

A. Structure Height Estimation

Structure height estimation has been an active area of research within the fields of remote sensing, computer vision, and photogrammetry. Earlier methods for height estimation have depended heavily on structurally and computationally expensive methods such as Light Detection and Ranging (LiDAR) and image stitching [15], [16]. As a result, cheaper methods have been proposed using various computer vision and photogrammetry techniques. Liu et al. [17] proposed Image to Elevation (IM2ELEVATION), which combined these techniques using CNNs to create accurate depth maps from public LiDAR point clouds. Despite outperforming other techniques on the International Society for Photogrammetry and Remote Sensing (ISRPS) Potsdam dataset, its lack of transferability makes it impractical. Chen et al. [18] proposed a Structure-Aware Building Height Estimation (SBHE) model that leverages a dual-branch deep learning architecture to better integrate structural constraints in height prediction. The technique showed promise for remote sensing-based urban analysis due to its high accuracy in urban morphology, but is limited by the quality of data required for accurate predictions. Eigen et al. [19] introduced one of the first deep learning approaches to depth estimation, using coarse global prediction with fine local refinements through a multi-scale CNN. However, its scalability is limited due to its dependence on ground-truth depth data.

B. Drone Datasets

Acquiring a diverse and high-resolution drone dataset is crucial for the development of an accurate semantic segmentation model. Previous large-scale datasets such as Pascal Visual Object Classes [20] and Microsoft Common Objects in Context [21] were utilized as benchmarks for contemporary drone datasets to perform various object recognition tasks. Following these datasets, the Cityscapes Dataset [22] provided 25,000 labeled images with 30 object classes. Additionally, the SYN-Seqs [23] dataset simulates oblique shots of vehicles and urban scenery in a similar fashion to our dataset. These baseline datasets made the synthesis of drone datasets possible, and remain fundamental research with applications in computer vision, autonomous driving, surveying, among other semantic segmentation tasks.

However, there are fewer UAV-based datasets composed of images captured at low-altitudes alongside diverse scenery. To the best of our knowledge, there are six drone datasets: Hyperspectral Imagery (HSI) [2], UAVid [3], Aeroscapes [4], Varied Drone Dataset (VDD) [5], Urban Drone Dataset (UDD) [6], and Institute of Computer Graphics and Vision (ICG) Drone Dataset [7]. HSI and Aeroscapes provide low-quality images, while the remaining have 4k or higher resolution images. UAVid and UDD both contain finely labelled imagery with intentions to assist 3D reconstruction using Structure from Motion (SfM) [24]. VDD presents diverse scenery as well as densely labelled images. Though these datasets provide rich and complex scenes, our research focuses solely on urban scenery. Additionally, an instrumental part of our parallax calculations is the known vertical displacement—a component not shown in any of the datasets. In our work, we introduce a basic amendable drone dataset centered around parallax geometry and distinct lines of sights of the same structure.

C. Parallax Method

Parallax—the changing of the displacement of an image from a different viewpoint—has long been used to estimate distances. First used in astronomy to calculate the distance between stars and still in use today [25], it is now employed in camera systems to measure depth and distance. Scharstein et al. [26] analyzed stereo correspondence algorithms that use two images to compute image displacements and disparity maps similar to parallax.

The study uses two images to determine the height of buildings by employing parallax and linearized pixel-to-angle mapping to reduce computational complexity. Similar studies have used linear mappings in single-shot distance estimation systems, such as the human-robot interface, which estimates distances using a linear relationship between the pixel angle mapping [27]. However, parallax has not been applied to estimate distances or building heights in the manner of this study.

III. Methodology

A. Data Collection

To identify the building outlines, our study aims to use deep learning segmentation on low-altitude UAV images which will be useful in estimating structure height using parallax geometry (detailed in Section III-D). Since there are no existing publicly available datasets that fit the purpose of this study with low-altitude UAV images paired with binary masks, we created a new custom building outline dataset. Our dataset contains 207 images collected from Google Maps Street View, each manually annotated using the Visual Geometry Group Image Annotator (VIA) [28] to create polygon outlines of the buildings. These polygons were later converted to masks to train the deep learning segmentation model. Fig. 1 shows a sample image in the dataset with its corresponding polygon overlay and generated mask.

Fig. 1. Sample image from the custom building outline dataset with manually annotated polygon overlay and final generated mask. Image © Google Street View.

B. Image Processing

To minimize noise and increase visibility of building edges, image processing techniques including Contrast Limited Adaptive Histogram Equalization (CLAHE) [29] and Laplacian of Gaussian (LoG) filter [30] were used. The before and after comparison is shown in Fig. 2.

CLAHE is a type of Histogram Equalization (HE) method that utilizes a clip limit to prevent the excessive amplification of noise often occurring when using HE, improving details in localized regions with varying lighting conditions. In the dataset used in this study, buildings images were taken from varying regions and weather conditions, making CLAHE useful for accounting for the varying lighting conditions across images. We implemented CLAHE using Open Computer Vision (OpenCV) with a clip limit of 2.0 and tile grid size of (8, 8).

The LoG filter comprises two image processing steps. First, Gaussian blur is applied to blur and smooth out the image to reduce noise, which includes unwanted details from surrounding objects like cars and trees. Next, a Laplacian filter is applied to amplify the precise edges of the buildings, making it easier to find the building. The Laplacian filter utilizes a method called zero crossings to detect edges by identifying where the Laplacian, or the second derivative of the image, changes sign. In our study, LoG was implemented using OpenCV functions: for Gaussian blur, a kernel size of (15, 15) and a SigmaX of 10 was used; for the Laplacian filter, a kernel size of 5 was used.

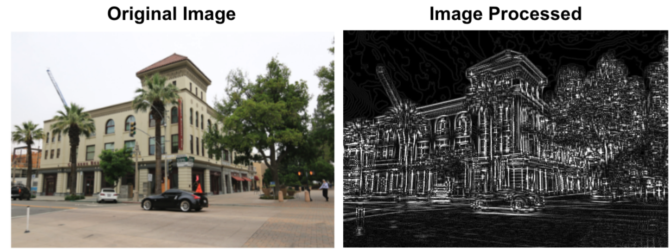

Fig. 2. Comparison of a sample image before and after applying the proposed image processing. Image © Google Street View.

C. Deep Learning-Based Image Segmentation

Our study leverages deep learning-based image segmentation to detect masks of buildings, which can be used to find their highest and lowest points. Identifying these points are important as they can be used for height estimation with parallax geometry, detailed in Section III-D. For this study, a DeepLabV3 model with a ResNet-50 backbone pre-trained on ImageNet is used. DeepLabV3 is a deep neural network for semantic segmentation tasks developed by researchers at Google, and it utilizes Atrous Spatial Pyramid Pooling (ASPP) to improve segmentation accuracy on objects with varying sizes. For this study, DeepLabV3 was implemented using Tensorflow Keras. Additional layers, including Batch Normalization and Dropout, were added to improve regularization. The Adam optimizer with a learning rate of 0.0001 was used. The ReduceLROnPlateau callback was used to halve the learning rate when performance plateaued for 8 epochs, and an Early Stopping callback was used with a patience of 15 epochs. Finally, a combined loss function consisting of Categorical Cross-entropy and validation accuracy was used. Standard performance metrics for segmentation models, including accuracy, Dice Coefficient (DC), and Intersection over Union (IoU) score, were used to evaluate performance.

D. Parallax-Based Height Estimation

To find the heights of our buildings, our study utilized a unique parallax technique. The overall methodology flow for calculating the building height using the drone is shown in Fig. 3. First, two images of a building are taken using the Da-Jiang Innovations (DJI) Mini 4K drone, one from a lower altitude and another from a higher altitude, after moving the camera vertically. As the drone user interface displays the vertical height of the drone, the heights of both images were recorded, which allows us to measure the vertical displacement.

The two images are then saved in a Secure Digital (SD) card and passed into the trained DeepLabV3 model, which outputs masks for each image, predicting the outline of the buildings at each point. Masks are generated by the model on a pixel-by-pixel basis, meaning that after mask generation, there is a possibility of numerous extraneous features that can impact height estimation. For this study, we implemented a Depth-First Search (DFS) algorithm in order to extract the largest feature by area. Each mask was generated as a black and white image. Connectivity between white pixels was denoted using an 8-connectivity criterion, labeling each contiguous region with a unique identifier. The area of each unique contiguous region was computed. All subsequent smaller areas had their pixel values set to the background pixel value (0), meaning only the largest contiguous region, typically intended to represent the outline of the building, remained. This post-processing step helps remove any spurious predictions and noise produced by the segmentation model.

With these cleaned masks, we used our proposed parallax height estimation method to compute the final building height. The calculation process, starting from the cleaned masks, is shown in Fig. 4. Throughout the process, the purple line segments indicate values that are known during that step from known drone specifications or calculated in previous steps. We begin by knowing the recorded vertical displacement and the drone camera specifications, which include a vertical resolution of $R_{\text{vertical}} = 2250$ pixels and a vertical Field of View ($\text{FOV}_{\text{vertical}} = 58.5°$).

We create two triangles, connecting the vertical displacement of the drone ($y_{\text{point,pixel}}$) to the top and bottom points of the building. To find the angles of each triangle we used a linear approximation where each individual pixel represents the vertical field of view divided by the vertical resolution, similar to the one used in [27] without the tilt. Using the known drone specifications, we can derive the angle θ between the horizontal line parallel to the ground from the camera and the line segment from the camera to the reference point using the following equation, shown in green in Fig. 4:

$$\theta = \frac{y_{\text{point,pixel}} - \frac{R_{\text{vertical}}}{2}}{R_{\text{vertical}}} \times \text{FOV}_{\text{vertical}} \tag{1}$$

This angle must be transformed to calculate angles of the constructed triangle. A 90° offset must be applied to find the angle from the surface of the camera to the point, creating the triangle in Fig. 4. The offset depends on the following cases: if this angle is the first frame and the reference point is the bottom point add 90° to the θ (angle A in Fig. 4); if it is the second frame for the bottom point, subtract the θ from 90° (angle B in Fig. 4); if it is the first frame for the top point, subtract the θ from 90°; if it is the second frame for the top point, add 90° to the θ. This is the most common configuration; however, if θ of the first frame bottom angle (angle A in Fig. 4) is negative, then the offsets applied to θ switch so it would be $90 - |\theta|$. If, for the second frame, the bottom angle is negative, then it would become $90 + |\theta|$. If, for the first frame, the top angle is positive, then it is $90 + |\theta|$, and if, for the second frame top angle is positive, it becomes $90 - |\theta|$. As two of the three angles for both triangles are now known, we find the third angle by subtracting the sum of the two angles from 180°. To solve for the other side lengths in the triangles, we use the law of sines, as shown in step 3 of Fig. 4. Finally, the building height is found by using the law of sines on any triangle connected to the side of the building, depicted by the green segment in step 5 of Fig. 4.

IV. Results and Discussion

A. DeepLabV3 Image Segmentation

The DeepLabV3 Image Segmentation model was assessed based on its accuracy in three metrics: validation accuracy, DC, and IoU. The manually annotated polygon masks served as ground truth and were compared to the predicted masks for each building across all three metrics. To evaluate the influence of preprocessing, we compared model performance on both processed and unprocessed images.

These results of the models are displayed in Fig. 5. Without image processing, our model had a maximum validation accuracy of 96.58% at epoch 24, a maximum validation DC of 96.19% at epoch 32, and a maximum validation IoU of 92.73% at epoch 32. With image processing, our model had a maximum validation accuracy of 95.95% at epoch 42, and maximum validation DC of 95.56% at epoch 47, and a maximum validation IoU of 91.52% at epoch 44. The unprocessed images yielded slightly higher peak performance across all three metrics. We hypothesize this could be due to amplified image noise created by processing of images. LoG and CLAHE filters applied on our images may have enhanced background artifacts in each image, degrading segmentation quality. Consequently, the model trained and tested on unprocessed images was used while estimating the height of University of California, Irvine (UCI) buildings.

B. Parallax Height Estimation on UCI Buildings

The parallax-based height estimation process involved implementing a DFS algorithm on each generated building mask, followed by the application of the parallax-based height estimation to each pair of cleaned masks. This yielded a single height estimation for each corresponding image pair. The process was conducted on the following seven buildings across the campus of UCI: Engineering Hall, Engineering Tower, Frederick Reines Hall, Interdisciplinary Science & Technology Building, John Croul Hall, Multipurpose Science & Technology Building, and Rowland Hall. The true values of the heights of these structures were obtained from architectural plans of the campus. In Table I, across the seven buildings analyzed, the average percent discrepancy between the true and estimated heights was 22.84%. However, three buildings—the Engineering Hall, John Croul Hall, and Rowland Hall—exhibited substantially higher percent discrepancies than the others, each due to distinct factors. For the Engineering Hall, the true height was measured from the front elevation, whereas the drone images of the building were taken from its backside where its elevation differs. This discrepancy was compounded by trees which also presented significant noise during mask generation. The Rowland Hall experienced similar challenges, as noise complicated the mask generation, and true height of

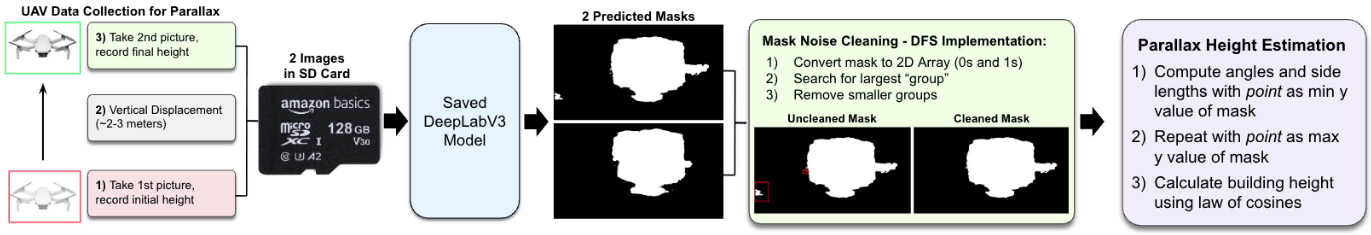

Fig. 3. Methodology flow for structure height estimation using UAV.

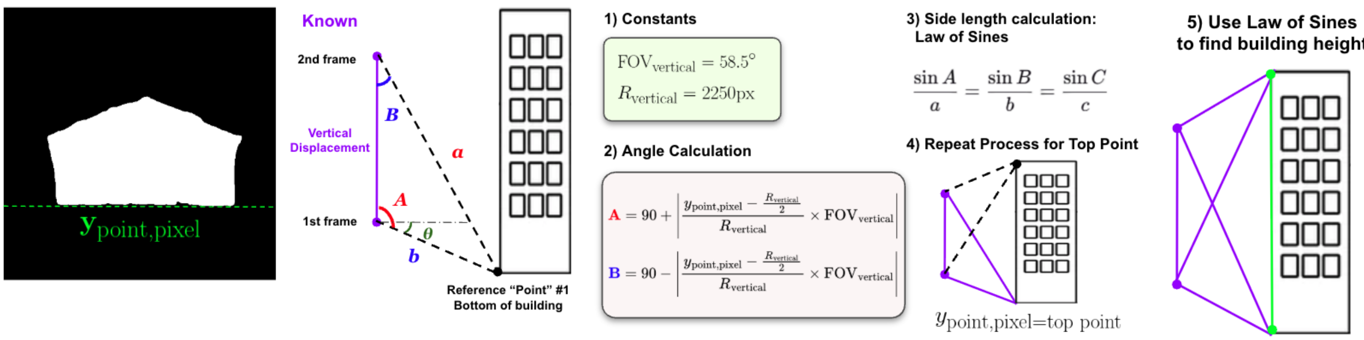

Fig. 4. Parallax height calculation process.

Fig. 5. Comparison of training and validation accuracy, DC, IoU, and validation loss with and without image processing.

the building may have been skewed by the sloping terrain around the building. The John Croul Hall exhibited the largest percent discrepancy, 91.86%, as shown in Table I, stemming from limitations with our DeepLabV3 segmentation model. The model was not trained to distinguish between adjacent structures. Since the John Croul Hall is a relatively short building directly adjacent to taller buildings behind it, our drone shots were ineffective in capturing the difference between the two buildings, leading our model to generate masks which included parts of other buildings not connected to the John Croul Hall, significantly overestimating the height relative to the ground truth. Excluding these three outliers, our model performed with an average percent discrepancy of 6.54% across the remaining buildings. The Engineering Tower had the lowest percent discrepancy at 1.60%, as shown in Table I, due to its isolated location away from surrounding noise and its simple, rectangular shape, which aligns well with our model's capabilities.

TABLE I
PERCENT DISCREPANCY BETWEEN GROUND TRUTH AND PREDICTED HEIGHTS

Building	**Percent Discrepancy (%)**
Engineering Hall	20.07
Engineering Tower	1.60
Frederick Reines Hall	9.31
Interdisciplinary Science & Engineering Building	6.75
John Croul Hall	91.86
Multipurpose Science & Technology Building	8.51
Rowland Hall	21.76

In order to test the precision and repeatability of height predictions, new masks were generated on the pair of images for the Engineering Tower twenty times. Due to the small sample size, a test statistic was computed using the Shapiro-Wilk Test to confirm normality [31].

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \tag{2}$$

Our calculated test statistic was $W = 0.9167$, obtaining a p-value of $p = 0.0855$. Since $p \geq 0.05$, we were able to assume normality and continue with analysis.

As shown in Fig. 6, the sample mean was $\bar{x} = 39.45\ m$ with a standard deviation of $\sigma = 2.26\ m$, meaning our proposed model was consistent in its height estimations. The high standard deviation is most likely due to a low sample size, and as more samples are taken the standard deviation is expected to decrease.

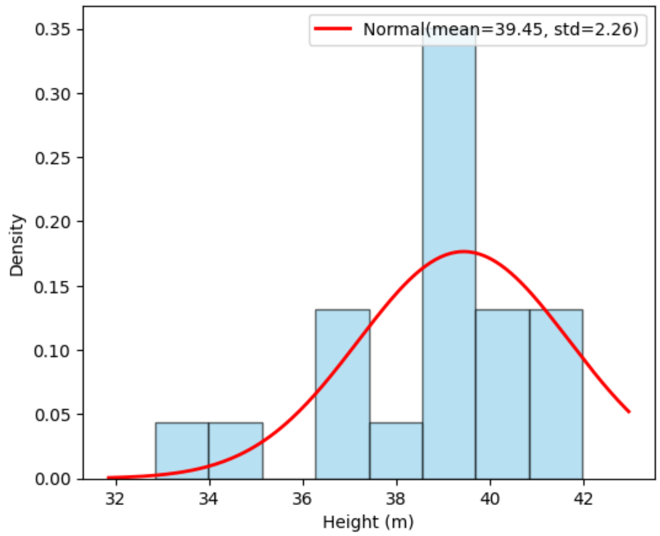

Fig. 6. Histogram of sample means from twenty height estimations of the UCI Engineering Tower.

C. Limitations and Future Directions

The proposed framework has several limitations: 1) The drone used could only record two significant figures in meters, limiting the accuracy of the vertical displacement between images, which ultimately causes inaccuracies in the estimated height. 2) The vFOV had to be estimated based on drone specifications, resulting in further possible discrepancies. 3) The deep learning model does not perform well on non-rectangular buildings and often catches background buildings as part of the main structure, changing the predicted pixel height of the top and bottom points of the building and resulting in some outcomes with 100% percent discrepancy. 4) Our model assumed that the whole building is in frame for both images, which can make it difficult to apply to scenarios where the building cannot be captured in frame easily. 5) Our model assumes the drone moves in a straight vertical line upwards from the first frame to the second frame, but the drone can sometimes move to the side due to wind. In the future, a drone with more precise height measurement and a known vFOV would improve results. A model trained on a broader set of images would also improve results on buildings with special shapes and buildings surrounded by unwanted artifacts like trees, cars, and background buildings. Lastly, future works could investigate whether it is possible to use this method to estimate building height without the whole building in frame by taking multiple images.

V. Conclusion

In this study, we combined DeepLabV3 instance segmentation models with parallax geometry-based height estimation to predict the heights of structures across the UCI campus. Our model achieved a maximum unprocessed DC of 96.19% and a processed DC of 95.56%, demonstrating strong performance in outlining building structures. The model showed reduced accuracy in scenarios with noisy images including people, vehicles, and background buildings. The heights had a 6.54% discrepancy when accounting for incorrect ground truth heights for the Engineering Hall, John Croul Hall, and Rowland Hall.

Accurate structure height estimation has significant applications in construction planning, evacuation management, and disaster response. Our work can be used to rapidly estimate the height of a given building from two drone shots, offering potential usability in emergencies such as building fires and earthquakes where first responders require precise measurements of structure heights. Future work could focus on improving model robustness against visual noise and expanding the dataset to encompass diverse environments.

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References

[1] G. Welch and G. Bishop, "An introduction to the Kalman filter," Univ. of North Carolina at Chapel Hill, Tech. Rep. TR 95-041, 1995.
[2] B. Niu, Q. Feng, B. Chen, C. Ou, Y. Liu, and J. Yang, "HSI-TransUNet: A transformer based semantic segmentation model for crop mapping from UAV hyperspectral imagery," *Comput. Electron. Agric.*, vol. 201, 2022, Art. no. 107297.
[3] Y. Lyu, G. Vosselman, G.-S. Xia, A. Yilmaz, and M. Y. Yang, "UAVid: A semantic segmentation dataset for UAV imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 165, pp. 108–119, 2020.
[4] I. Nigam, C. Huang, and D. Ramanan, "Ensemble knowledge transfer for semantic segmentation," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2018, pp. 1499–1508.
[5] W. Cai, K. Jin, J. Hou, C. Guo, L. Wu, and W. Yang, "VDD: Varied drone dataset for semantic segmentation," *J. Vis. Commun. Image Represent.*, vol. 109, 2025, Art. no. 104429.
[6] Y. Chen, Y. Wang, P. Lu, Y. Chen, and G. Wang, "Large-scale structure from motion with semantic constraints of aerial images," in *Pattern Recognition and Computer Vision (PRCV 2018)*, Lecture Notes in Computer Science, vol. 11256. Cham, Switzerland: Springer, 2018, pp. 347–358.
[7] B. Sari and Graz University of Technology, "Semantic Drone Dataset," [Online]. Available: https://www.kaggle.com/datasets/bulentsiyah/semantic-drone-dataset.
[8] A. Gupta, M. P. M. Merchant, U. Verma, and R. M. Pai, "Semantic segmentation of UAV aerial videos using convolutional neural networks," in *Proc. IEEE Int. Conf. Artif. Intell. Knowl. Eng. (AIKE)*, 2019, pp. 21–27.
[9] M. Tang, K. Georgiou, H. Qi, C. Champion, and M. Bosch, "Semantic segmentation in aerial imagery using multi-level contrastive learning with local consistency," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2023, pp. 3787–3796.
[10] M. A. Perryman, L. Lindegren, J. Kovalevsky, E. Hoeg, U. Bastian, P. L. Bernacca, *et al.*, "The HIPPARCOS catalogue," *Astron. Astrophys.*, vol. 323, pp. L49–L52, 1997.
[11] A. G. Brown, A. Vallenari, T. Prusti, J. H. J. de Bruijne, C. Babusiaux, C. A. Bailer-Jones, *et al.*, "Gaia data release 2: Summary of the contents and survey properties," *Astron. Astrophys.*, vol. 616, Art. no. A1, 2018.
[12] A. G. Riess, S. Casertano, W. Yuan, L. Macri, J. Anderson, J. W. MacKenty, *et al.*, "New parallaxes of galactic cepheids from spatially scanning the Hubble Space Telescope: Implications for the Hubble constant," *Astrophys. J.*, vol. 855, no. 2, Art. no. 136, 2018.

[13] G. F. Benedict, B. E. McArthur, M. W. Feast, T. G. Barnes, T. E. Harrison, R. J. Patterson, *et al.*, "Hubble Space Telescope fine guidance sensor parallaxes of Galactic Cepheid variable stars: Period-luminosity relations," *Astron. J.*, vol. 133, no. 4, pp. 1810–1827, 2007.
[14] F. Nex and F. Remondino, "UAV for 3D mapping applications: A review," *Appl. Geomatics*, vol. 6, no. 1, pp. 1–15, 2014.
[15] M. Mielcarek, K. Stereńczak, and A. Khosravipour, "Testing and evaluating different LiDAR-derived canopy height model generation methods for tree height estimation," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 71, pp. 132–143, 2018.
[16] W. Cai, S. Du, and W. Yang, "UAV image stitching by estimating orthograph with RGB cameras," *J. Vis. Commun. Image Represent.*, vol. 94, 2023, Art. no. 103835.
[17] C. J. Liu, V. A. Krylov, P. Kane, G. Kavanagh, and R. Dahyot, "IM2ELEVATION: Building height estimation from single-view aerial imagery," *Remote Sens.*, vol. 12, no. 17, 2020, Art. no. 2719.
[18] Y. Chen, J. Zhou, C. Xu, Q. Ma, X. Zhang, Y. N. Zhou, and Y. Ge, "Structure-aware deep learning network for building height estimation," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 130, 2025, Art. no. 104443.
[19] D. Eigen, C. Puhrsch, and R. Fergus, "Depth map prediction from a single image using a multi-scale deep network," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 27, 2014.
[20] M. Everingham, S. Eslami, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The PASCAL visual object classes challenge: A retrospective," *Int. J. Comput. Vis.*, vol. 111, no. 1, pp. 98–136, Jan. 2015.
[21] T. Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Zurich, Switzerland, 2014, pp. 740–755.
[22] M. Cordts, M. Omran, S. Ramos, T. Rehfeld, M. Enzweiler, R. Benenson, U. Franke, S. Roth, and B. Schiele, "The Cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 3213–3223.
[23] G. Ros, L. Sellart, J. Materzynska, D. Vázquez, and A. M. López, "The SYNTHIA dataset: A large collection of synthetic images for semantic segmentation of urban scenes," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 3234–3243.
[24] J. L. Schönberger and J.-M. Frahm, "Structure-From-Motion Revisited," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 4104–4113.
[25] M. Andriantsaralaza, S. Ramstedt, W. H. T. Vlemmings, and E. De Beck, "Distance estimates for AGB stars from parallax measurements," *Astron. Astrophys.*, vol. 667, Art. no. A74, Nov. 2022.
[26] D. Scharstein, R. Szeliski, and R. Zabih, "A taxonomy and evaluation of dense two-frame stereo correspondence algorithms," in *Proc. IEEE Workshop Stereo and Multi-Baseline Vision (SMBV)*, 2001.
[27] B. A. Maxwell, N. Ward, and F. Heckel, "A human-robot interface for urban search and rescue," in *Proc. AAAI Mobile Robot Competition*, vol. 3, no. 1, 2003.
[28] Visual Geometry Group (VGG), "VGG image annotator (VIA)," *VGG Project / Official Statistics Annotate*, [Online]. Available: https://annotate.officialstatistics.org/.
[29] K. Zuiderveld, "Contrast limited adaptive histogram equalization," in *Graphics Gems IV*, P. S. Heckbert, Ed. San Diego, CA: Academic Press Professional, 1994, pp. 474–485.
[30] D. Marr and E. Hildreth, "Theory of edge detection," *Proceedings of the Royal Society of London. Series B. Biological Sciences*, vol. 207, no. 1167, pp. 187–217, 1980.
[31] S. S. Shapiro and M. B. Wilk, "An analysis of variance test for normality (complete samples)," *Biometrika*, vol. 52, no. 3/4, pp. 591–611, 1965.